Constant Maturity Treasury Convexity Correction

Working notes on Pucci (2014)

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1 Trade and Motivation

A *Constant Maturity Yield* (CMY) swap pays, on its exotic leg, the yield-to-maturity (i.e. internal rate of return) of a reference bond. When the reference is a par bond on a government yield curve at a fixed maturity (e.g. 10Y), the index is termed *Constant Maturity Treasury* (CMT). CMT indices are published daily by the Federal Reserve (US) and the Agence France Trésor (France); the Italian Rendistato is a similar bond-yield index. Their primary use is the indexing of ARM mortgages and the issuance of CMT-linked floating notes by financial institutions.

A note holder receiving the CMT benefits from a *worsening* of the referenced government's credit – the price-yield relationship makes the yield go up when the bond cheapens. This is the credit dimension that classical CMS pricing overlooks.

Pricing problem. The CMT-let pays at T_p the rate $R^{\mathrm{cmt}}_{T_s}$ fixed at T_s. As for any constant maturity product, the price involves a convexity adjustment beyond the bare forward yield. The textbook formula (Hull) approximates this via a first-order expansion of the yield-to-price relationship and *ignores*:

- default risk of the issuer of the reference bond curve;
- the impact of smile on the forward CMT;
- the interaction between default events and payment-date conventions.

Modelling goal. Build a credit-aware framework for CMT pricing that:

- recovers the classical Pelsser/Hagan formula in the no-default limit;
- handles three natural default-conditioned payoff variants (A, B, C) corresponding to different settlement conventions at default;
- delivers analytical convexity corrections under standard assumptions (independence of default time and rate filtration, plus absolute continuity of the hazard process);
- extends to smile via static replication on a risky CMS smile.

2 Yield-to-Maturity and Constant Maturity Treasury

2.1 YTM definition

Let $T_s < T_1 < \cdots < T_n$ be the bond's payment schedule with year fractions $y_i = \mathrm{yf}(T_{i-1}, T_i)$, fixed coupon rate K, and price process $(B_t)_{t\geq 0}$. The *yield to maturity* (YTM) at the reset date T_s is the rate $R^{\mathrm{ytm}}_{T_s}$ such that

$$B_{T_s} \;=\; \prod_{i=1}^{n} \frac{1}{1 + y_i\, R^{\mathrm{ytm}}_{T_s}} + K \sum_{i=1}^{n} \prod_{j=1}^{i} \frac{1}{1 + y_j\, R^{\mathrm{ytm}}_{T_s}}. \tag{1}$$

For $t \in (T_s, T_1)$ the formula is generalised by a stub-period adjustment with exponent $\frac{T_1 - t}{T_1 - T_s}$ on the first factor.

Annual 30/360 simplification. Using the algebraic identity $(1+a)^{-n} + \sum_{i=1}^{n} a/(1+a)^i = 1$, for an annual bond with standard 30/360 conventions (1) reduces to the textbook form

$$B_{T_s} \;=\; (1 + R^{\mathrm{ytm}}_{T_s})^{-n} + \frac{K}{R^{\mathrm{ytm}}_{T_s}}\big(1 - (1 + R^{\mathrm{ytm}}_{T_s})^{-n}\big). \tag{2}$$

The YTM is a non-linear function of the bond price, so YTM-contingent derivatives do not generally admit closed forms.

2.2 CMT as par yield of the curve

The spot CMT is normally computed by interpolating internal rates of return of bonds with expiries closest to the prescribed tenor. To sidestep the noise from off-the-run bonds, we model the CMT directly through fundamental building blocks of the (defaultable) bond curve:

Definition 1 (Constant Maturity Treasury)**.** *The CMT $R^{\mathrm{cmt}}_{T_s} = R^{\mathrm{cmt}}_{T_s}(T_1, \ldots, T_n)$ is the coupon rate that values the (defaultable) bond at par at T_s:*

$$1 \;=\; R^{\mathrm{cmt}}_{T_s}\, \widehat{\widehat{A}}_{T_s} \;+\; \widehat{\widehat{D}}_{T_s T_n} \;+\; \mathrm{Rec}\,[\textit{default leg}]_{T_s}, \tag{3}$$

where $\widehat{\widehat{D}}, \widehat{\widehat{A}}$ are the credit-risk-adjusted discount and annuity defined in Section 3, and Rec *is the recovery rate on the accelerated principal.*

2.3 The YTM–CMT bridge

Proposition 1 (YTM–CMT Taylor expansion)**.** *The YTM and CMT are linked by*

$$\boxed{R^{\mathrm{ytm}}(B, K) \;\cong\; R^{\mathrm{cmt}} + (K - R^{\mathrm{cmt}}) \;-\; \frac{R^{\mathrm{cmt}}}{1 - (1 + R^{\mathrm{cmt}})^{-n}}\,(B - 1).} \tag{4}$$

For large n, the prefactor approaches $1/n$, giving the simpler form $R^{\mathrm{ytm}} \cong R^{\mathrm{cmt}} + (K - R^{\mathrm{cmt}}) - (B-1)/n$.

Proof. Define $f(R, K) \equiv (1+R)^{-n} + (K/R)(1 - (1+R)^{-n})$, the right-hand side of (2). Then $R^{\mathrm{ytm}}(B, K)$ is implicitly defined by $f(R^{\mathrm{ytm}}(B, K), K) = B$. By construction $f(R^{\mathrm{cmt}}, R^{\mathrm{cmt}}) = 1$, hence $R^{\mathrm{ytm}}(1, R^{\mathrm{cmt}}) = R^{\mathrm{cmt}}$.

Computing partials:

$$\begin{aligned}
\partial_K f \;&=\; \frac{1 - (1+R)^{-n}}{R}, \\
\partial_R f \;&=\; n\left(\frac{K}{R} - 1\right)(1+R)^{-n-1} - \frac{K}{R^2}\big(1 - (1+R)^{-n}\big).
\end{aligned}$$

Implicit differentiation of $f(R^{\mathrm{ytm}}(B,K),K)=B$:

$$\partial_R f\,\partial_B R^{\mathrm{ytm}}=1,\qquad \partial_R f\,\partial_K R^{\mathrm{ytm}}+\partial_K f=0.$$

At $R=K=R^{\mathrm{cmt}}$, $(K/R-1)=0$ kills the first term of $\partial_R f$ and $K/R^2=1/R^{\mathrm{cmt}}$, so $\partial_R f(R^{\mathrm{cmt}},R^{\mathrm{cmt}})=-((1-(1+R^{\mathrm{cmt}})^{-n})/R^{\mathrm{cmt}})$. Hence

$$\partial_B R^{\mathrm{ytm}}\big|_{(1,R^{\mathrm{cmt}})}=\frac{-R^{\mathrm{cmt}}}{1-(1+R^{\mathrm{cmt}})^{-n}}\cdot(-1)=\frac{R^{\mathrm{cmt}}}{(1+R^{\mathrm{cmt}})^{-n}-1},\quad \partial_K R^{\mathrm{ytm}}\big|_{(1,R^{\mathrm{cmt}})}=-\frac{\partial_K f}{\partial_R f}=1.$$

The first-order Taylor expansion of R^{ytm} about $(B,K)=(1,R^{\mathrm{cmt}})$ yields the result. □

The bridge justifies treating R^{cmt} as a tractable proxy for R^{ytm} in pricing, with the Taylor remainder controlling the approximation error.

3 Modelling Framework: Credit-Risk-Adjusted Quantities

3.1 Filtrations and the hazard process

Following Bielecki–Jeanblanc–Rutkowski [1], Jeanblanc–Rutkowski [2], Jamshidian [3], and Brigo–Morini [4]:

- τ is the default time (a positive random variable),
- $\mathcal{F}=(\mathcal{F}_t)_{t\geq 0}$ is the *default-free* filtration (no default information),
- $\mathcal{G}=(\mathcal{G}_t)_{t\geq 0}$ is the *market* filtration $\mathcal{G}_t=\mathcal{F}_t\vee\sigma(\tau\wedge t)$ (the smallest filtration making τ a $\mathcal{G}$-stopping time and containing $\mathcal{F}$),
- the hazard process is $\Gamma_t\equiv-\log\mathbb{P}^N[\tau>t\mid\mathcal{F}_t]$, so that $e^{-\Gamma_t}=\mathbb{P}^N[\tau>t\mid\mathcal{F}_t]$ is the Azéma supermartingale [5],
- $F_t\equiv 1-e^{-\Gamma_t}$ is an increasing continuous process.

3.2 Risky and credit-risk-adjusted discount bonds

Definition 2 (Three discount bonds)**.** *Under any martingale measure $\mathbb{Q}^N$ with numeraire N_t,*

$$D_{tT} = N_t\,\mathbb{E}^N\left[\frac{1}{N_T}\,\middle|\,\mathcal{F}_t\right] \qquad \textit{(risk-free, } \mathcal{F}_t\textit{-conditional)} \tag{5}$$

$$\widehat{D}_{tT} = N_t\,\mathbb{E}^N\left[\frac{\mathbf{1}_{\tau>T}}{N_T}\,\middle|\,\mathcal{G}_t\right] \qquad \textit{(risky, } \mathcal{G}_t\textit{-conditional)} \tag{6}$$

$$\widehat{\widehat{D}}_{tT} = N_t\,\mathbb{E}^N\left[\frac{e^{\Gamma_t-\Gamma_T}}{N_T}\,\middle|\,\mathcal{F}_t\right] \qquad \textit{(credit-risk-adjusted, } \mathcal{F}_t\textit{-conditional)} \tag{7}$$

The credit-risk-adjusted (CRA) bond admits the equivalent forms

$$\widehat{\widehat{D}}_{tT} = D_{tT}\,\mathbb{E}^T\left[e^{\Gamma_t-\Gamma_T}\,\middle|\,\mathcal{F}_t\right] = D_{tT}\,e^{\Gamma_t}\,\mathbb{P}^T[\tau>T\mid\mathcal{F}_t], \tag{8}$$

under the T-forward measure. The link between the risky and CRA bonds is

$$\widehat{D}_{tT} = \mathbf{1}_{\tau>t}\,\widehat{\widehat{D}}_{tT},\qquad \widehat{\widehat{D}}_{tT}>0 \text{ a.s.} \tag{9}$$

Why CRA? The CRA quantities are positive even after default and avoid the singularity that $\widehat{D}_{tT}/\widehat{D}_{tT}$-type ratios exhibit at τ. This is what makes the modelling tractable.

3.3 Risky annuity and risky swap rates

Definition 3 (CRA risky annuity and swap rates)**.**

$$\widehat{\widehat{A}}_t = \sum_{i=1}^{n} y_i \, \widehat{\widehat{D}}_{tT_i}, \tag{10}$$

$$\widehat{\widehat{R}}_t^{\text{swp}} \equiv \frac{\widehat{\widehat{D}}_{tT_s} - \widehat{\widehat{D}}_{tT_n}}{\widehat{\widehat{A}}_t} \qquad \textit{(recoveryless)}, \tag{11}$$

$$\widehat{R}_t^{\text{swp}} \equiv \frac{\widehat{\widehat{D}}^{(R)}_{tT_s} - \widehat{\widehat{D}}^{(R)}_{tT_n}}{\widehat{\widehat{A}}_t} \qquad \textit{(with recovery)}, \tag{12}$$

where $\widehat{\widehat{D}}^{(R)}_{tT} = \widehat{\widehat{D}}_{tT} + \text{Rec}\,[\textit{default leg}]_t$ *includes a deterministic recovery* Rec *paid at default.*

Proposition 2 (CMT is the risky-with-recovery swap rate)**.** $R_t^{\text{cmt}} = \widehat{R}_t^{\text{swp}}$ *for all* $t \le T_s$.

Proof. Equation (3) states that $R_{T_s}^{\text{cmt}}$ values the (defaultable, with-recovery) bond at par at T_s: $1 = R_{T_s}^{\text{cmt}} \widehat{\widehat{A}}_{T_s} + \widehat{\widehat{D}}^{(R)}_{T_sT_n}$, equivalently $\widehat{\widehat{A}}_{T_s} R_{T_s}^{\text{cmt}} = 1 - \widehat{\widehat{D}}^{(R)}_{T_sT_n}$. The right-hand side is exactly $\widehat{\widehat{D}}^{(R)}_{T_sT_s} - \widehat{\widehat{D}}^{(R)}_{T_sT_n}$ since $\widehat{\widehat{D}}^{(R)}_{T_sT_s} = 1$. Comparing with (12) gives $R_{T_s}^{\text{cmt}} = \widehat{R}_{T_s}^{\text{swp}}$. The $t < T_s$ identification follows by the same construction at the forward level. □

This identification is the central conceptual move: *the CMT is a risky-with-recovery forward swap rate*, hence – as a ratio of tradables – it is a martingale under the natural numeraire measure.

Corollary 1 (CMT martingale property)**.** *Under the* $\widehat{\widehat{A}}_t$*-numeraire (risky annuity) measure* $\mathbb{Q}^{\widehat{\widehat{A}}}$, R_t^{cmt} *is an* $\mathcal{F}$*-martingale.*

3.4 Standing assumptions

Assumption 1 (DI – Default Independence)**.** τ *is independent of* $\mathcal{F}$.

Assumption 2 (AC – Absolute Continuity)**.** $F_t = 1 - e^{-\Gamma_t}$ *is absolutely continuous, hence* $\Gamma_t = \int_0^t \gamma_u \, du$ *for some intensity process* $\gamma_t \ge 0$.

Under DI: $\widehat{\widehat{D}}_{tT} = D_{tT} \, e^{\Gamma_t - \Gamma_T}$. Under DI+AC: $\mathbb{P}^T[\tau > T \mid \mathcal{F}_T] = e^{-\int_0^T \gamma_u du}$ with γ deterministic.

4 The Three Payoff Versions

For a CMT-let / YTM-let, the convention regarding what happens at default of the reference issuer matters. We consider three natural variants, denoting R either R^{cmt} or R^{ytm}:

Definition 4 (Three variants of the CMT/YTM-let)**.** *At payment date* $T_p \ge T_s$*:*

(A) Survival to payment. *Pays* $\mathbf{1}_{\tau > T_p} R_{T_s}$. *The coupon vanishes if default occurs at any time before payment date.*

(B) Survival to fix. *Pays* $\mathbf{1}_{\tau > T_s} R_{T_s}$. *The coupon vanishes only if default precedes the fixing date; default between* T_s *and* T_p *does not extinguish the payment.*

(C) Fix-or-pre-default. *Pays* $Y \equiv \mathbf{1}_{\tau > T_s} R_{T_s} + \mathbf{1}_{\tau \le T_s} R_{\tau -}$*, i.e. if default occurs before* T_s*, the holder receives the last pre-default fixing of the rate.*

By construction $\text{NPV}^{(C)} \ge \text{NPV}^{(B)} \ge \text{NPV}^{(A)}$. Variants (A) and (B) typically embed in survival-contingent swaps (e.g. a vanilla leg that cancels at default), while (C) is the building block of non-contingent CMT-linked structures.

4.1 Convexity correction

Definition 5 (Convexity correction toolkit)**.**

$$\mathrm{CC}^{(i)} \;\equiv\; \frac{\mathrm{NPV}^{(i)}}{D \cdot F} - 1, \qquad i \in \{A, B, C\}, \tag{13}$$

where F is the relevant forward rate (here R^{cmt}) and D is the "natural" discount factor for the claim. The product $D \cdot F$ is the value of $\mathrm{NPV}^{(i)}$ when the rate volatility is set to zero.

5 Pricing Framework Identities

5.1 Conditional expectation lemmas

Lemma 1 (Survival-to-payment expectation, version A)**.** *For any $X \in \mathcal{F}_{T_s}$,*

$$\mathbb{E}^N\left[\frac{X\,\mathbf{1}_{\tau>T_p}}{N_{T_p}}\,\middle|\,\mathcal{G}_t\right] \;=\; \mathbf{1}_{\tau>t}\,\mathbb{E}^N\left[X\,e^{\Gamma_t-\Gamma_{T_s}}\,\frac{\widehat{\widehat{D}}_{T_sT_p}}{N_{T_s}}\,\middle|\,\mathcal{F}_t\right]. \tag{14}$$

Proof. Following Jeanblanc–Rutkowski [2] Corollary 2.6 (or Jamshidian [3] Theorem 3.5):

$$\begin{aligned}
\mathbb{E}^N\left[\frac{X\,\mathbf{1}_{\tau>T_p}}{N_{T_p}}\,\middle|\,\mathcal{G}_t\right] &= \mathbb{E}^N\left[X\,\mathbb{E}^N\left[\frac{\mathbf{1}_{\tau>T_p}}{N_{T_p}}\,\middle|\,\mathcal{G}_{T_s}\right]\,\middle|\,\mathcal{G}_t\right] \\
&= \mathbb{E}^N\left[X\,\mathbf{1}_{\tau>T_s}\,\mathbb{E}^N\left[\frac{e^{\Gamma_{T_s}-\Gamma_{T_p}}}{N_{T_p}}\,\middle|\,\mathcal{F}_{T_s}\right]\,\middle|\,\mathcal{G}_t\right] \\
&= \mathbb{E}^N\left[X\,\mathbf{1}_{\tau>T_s}\,\frac{\widehat{\widehat{D}}_{T_sT_p}}{N_{T_s}}\,\middle|\,\mathcal{G}_t\right] \\
&= \mathbf{1}_{\tau>t}\,\mathbb{E}^N\left[X\,e^{\Gamma_t-\Gamma_{T_s}}\,\frac{\widehat{\widehat{D}}_{T_sT_p}}{N_{T_s}}\,\middle|\,\mathcal{F}_t\right],
\end{aligned}$$

using the standard identity that splits a $\mathcal{G}$-conditional expectation of $\mathbf{1}_{\tau>T_s}\cdot\mathcal{F}_{T_s}$-measurable content into $\mathbf{1}_{\tau>t}\,e^{\Gamma_t-\Gamma_{T_s}}$ times an $\mathcal{F}$-conditional expectation. □

Lemma 2 (Survival-to-fix expectation, version B)**.** *For any $X \in \mathcal{F}_{T_s}$,*

$$\mathbb{E}^N\left[\frac{X\,\mathbf{1}_{\tau>T_s}}{N_{T_p}}\,\middle|\,\mathcal{G}_t\right] \;=\; \mathbf{1}_{\tau>t}\,\mathbb{E}^N\left[X\,e^{\Gamma_t-\Gamma_{T_s}}\,\frac{D_{T_sT_p}}{N_{T_s}}\,\middle|\,\mathcal{F}_t\right]. \tag{15}$$

Proof. Same recipe: condition on $\mathcal{G}_{T_s}$, use $\mathbf{1}_{\tau>T_s}/N_{T_p} = \mathbf{1}_{\tau>T_s}\,D_{T_sT_p}/N_{T_s}$ (the risk-free discount enters because the survival indicator is set at T_s), then collapse $\mathcal{G}_t$ to $\mathcal{F}_t$ via the same hazard-process identity. □

5.2 NPV expressions

Applying Lemmas 1–2 to $X = R_{T_s}$:

$$\mathrm{NPV}_t^{(A)} \;=\; \mathbf{1}_{\tau>t}\,N_t\,\mathbb{E}^N\left[R_{T_s}\,e^{\Gamma_t-\Gamma_{T_s}}\,\frac{\widehat{\widehat{D}}_{T_sT_p}}{N_{T_s}}\,\middle|\,\mathcal{F}_t\right], \tag{16}$$

$$\mathrm{NPV}_t^{(B)} \;=\; \mathbf{1}_{\tau>t}\,N_t\,\mathbb{E}^N\left[R_{T_s}\,e^{\Gamma_t-\Gamma_{T_s}}\,\frac{D_{T_sT_p}}{N_{T_s}}\,\middle|\,\mathcal{F}_t\right]. \tag{17}$$

For variant (C), recall the decomposition $Y = \mathbf{1}_{\tau>T_s}R_{T_s} + \mathbf{1}_{\tau\le T_s}R_{\tau-}$:

Lemma 3 (Variant C decomposition)**.**

$$\mathrm{NPV}_t^{(C)} \;=\; \mathrm{NPV}_t^{(B)} \;+\; \mathbf{1}_{\tau>t}\, N_t\, \mathbb{E}^N\!\left[\int_t^{T_s} e^{\Gamma_t-\Gamma_u}\, R_u\, \frac{D_{uT_p}}{N_u}\, d\Gamma_u \,\middle|\, \mathcal{F}_t\right]. \tag{18}$$

Proof. The (B) piece is identical. For the second piece (default before T_s),

$$N_t\, \mathbb{E}^N\!\left[\frac{\mathbf{1}_{t<\tau\le T_s}\, R_\tau\, D_{\tau T_p}}{N_\tau} \,\middle|\, \mathcal{G}_t\right]$$

is a stochastic integral against the default-leg measure $d\Gamma$ on $[t, T_s]$. Using Jeanblanc–Rutkowski [2] Corollary 2.6 (or Jamshidian [3] Corollary 3.7), the integrand at τ maps to its $\mathcal{F}$-predictable counterpart $R_u\, D_{uT_p}/N_u$ multiplied by the hazard differential $e^{\Gamma_t-\Gamma_u}\, d\Gamma_u$. Continuity of R ensures $R_{\tau-} = R_\tau$ a.s. □

Picking $N_t = \widehat{\widehat{A}}_t$ makes R_t^{cmt} an $\mathcal{F}$-martingale by Corollary 1, so we work under $\mathbb{Q}^{\widehat{\widehat{A}}}$ from now on.

6 One-Factor Linear Risky Swap-Rate Model

6.1 Linear assumption

Assumption 3 (Linear risky swap-rate model)**.** *For every $U \in \{T_p, T_1, \ldots, T_n\}$,*

$$\frac{\widehat{\widehat{D}}_{T_sU}}{\widehat{\widehat{A}}_{T_s}} \;=\; G_U(R_{T_s}^{\mathrm{cmt}}), \qquad G_U(x) = \alpha + \beta_U\, x. \tag{19}$$

This generalises the Hagan [6] CMS linear-swap-rate assumption to the risky setting – the CMT being itself a risky swap rate (Proposition 2).

6.2 Calibration lemma

Define $\chi_t \equiv e^{\Gamma_{T_s}-\Gamma_t}$ (a deterministic function of t under DI). Note $\chi_{T_s} = 1$.

Lemma 4 (Forward CRA-discount-to-annuity ratio)**.** *Under DI,*

$$\mathbb{E}^{\widehat{\widehat{A}}}\!\left[\frac{\widehat{\widehat{D}}_{T_sU}}{\widehat{\widehat{A}}_{T_s}} \,\middle|\, \mathcal{F}_t\right] \;=\; \chi_t\, \frac{\widehat{\widehat{D}}_{tU}}{\widehat{\widehat{A}}_t}. \tag{20}$$

Proof. Using the tower property and the closed form $\widehat{\widehat{D}}_{tU} = D_{tU}\, e^{\Gamma_t-\Gamma_U}$ (under DI):

$$\begin{aligned}
\mathbb{E}^{\widehat{\widehat{A}}}\!\left[\frac{\widehat{\widehat{D}}_{T_sU}}{\widehat{\widehat{A}}_{T_s}} \,\middle|\, \mathcal{F}_t\right] &= \mathbb{E}^{\widehat{\widehat{A}}}\!\left[\mathbb{E}^{\widehat{\widehat{A}}}\!\left[\frac{e^{\Gamma_{T_s}-\Gamma_U}}{\widehat{\widehat{A}}_U} \,\middle|\, \mathcal{F}_{T_s}\right] \,\middle|\, \mathcal{F}_t\right] \\
&= e^{\Gamma_{T_s}-\Gamma_U}\, \mathbb{E}^{\widehat{\widehat{A}}}\!\left[\frac{D_{UU}}{\widehat{\widehat{A}}_U} \,\middle|\, \mathcal{F}_t\right] \qquad (D_{UU} = 1) \\
&= e^{\Gamma_{T_s}-\Gamma_U}\, \frac{D_{tU}}{\widehat{\widehat{A}}_t} \;=\; e^{\Gamma_{T_s}-\Gamma_t}\, \frac{D_{tU}\, e^{\Gamma_t-\Gamma_U}}{\widehat{\widehat{A}}_t} \;=\; \chi_t\, \frac{\widehat{\widehat{D}}_{tU}}{\widehat{\widehat{A}}_t}.
\end{aligned}$$

□

Proposition 3 (Calibration of α, β_U)**.** *Under DI and Assumption 3,*

$$\alpha \;=\; \frac{1}{\sum_{i=1}^{n} y_i}, \qquad \beta_U \;=\; \frac{\chi_0\, \widehat{\widehat{D}}_{0U}/\widehat{\widehat{A}}_0 - \alpha}{R_0^{\mathrm{cmt}}}. \tag{21}$$

Proof. For *any* R_0^{cmt}, the algebraic identity $\sum y_i \widehat{\widehat{D}}_{0T_i} = \widehat{\widehat{A}}_0$ rearranges to

$$1 \;=\; \sum_i y_i \frac{\widehat{\widehat{D}}_{0T_i}}{\widehat{\widehat{A}}_0} \;=\; \alpha\left(\sum_i y_i\right) + R_0^{\mathrm{cmt}} \sum_i y_i\, \beta_{T_i}, \tag{22}$$

using Lemma 4 at $t = 0$ together with (19). Sending $R_0^{\mathrm{cmt}} \to 0$ yields $\alpha = 1/\sum y_i$. The expression for β_U follows from Lemma 4 and (19): $\alpha + \beta_U R_0^{\mathrm{cmt}} = \chi_0 \widehat{\widehat{D}}_{0U}/\widehat{\widehat{A}}_0$. □

Corollary 2 (Pre-fix CRA-discount-to-annuity ratio)**.** *For $t \le T_s$,*

$$\frac{\widehat{\widehat{D}}_{tU}}{\widehat{\widehat{A}}_t} \;=\; \frac{G_U(R_t^{\mathrm{cmt}})}{\chi_t} \;=\; \frac{\alpha + \beta_U\, R_t^{\mathrm{cmt}}}{\chi_t}. \tag{23}$$

6.3 NPV reductions

Substituting (19) into the NPV expressions:

Proposition 4 (NPV reductions under DI + linear model)**.**

$$\mathrm{NPV}_t^{(A)} \;=\; \mathbf{1}_{\tau>t}\, \widehat{\widehat{A}}_t\, e^{\Gamma_t - \Gamma_{T_s}}\, \mathbb{E}^{\widehat{\widehat{A}}}\Big[R_{T_s}^{\mathrm{cmt}}\big(\alpha + \beta_{T_p} R_{T_s}^{\mathrm{cmt}}\big) \,\Big|\, \mathcal{F}_t \Big], \tag{24}$$

$$\mathrm{NPV}_t^{(B)} \;=\; e^{\Gamma_{T_p} - \Gamma_{T_s}}\, \mathrm{NPV}_t^{(A)}. \tag{25}$$

Proof. For (A), apply (19) with $U = T_p$ to the integrand of (16); the $\mathbb{E}$ becomes the stated form. For (B), use $D_{T_sT_p} = \widehat{\widehat{D}}_{T_sT_p}\, e^{\Gamma_{T_p} - \Gamma_{T_s}}$ under DI and AC to obtain the relation (25). □

For (C), Lemma 3 requires evaluating a stochastic integral against $d\Gamma$. Splitting via Corollary 2:

$$Q \;\equiv\; \mathbf{1}_{\tau>t}\, \widehat{\widehat{A}}_t\, \mathbb{E}^{\widehat{\widehat{A}}}\left[\int_t^{T_s} e^{\Gamma_t - \Gamma_u}\, R_u^{\mathrm{cmt}}\, \frac{D_{uT_p}}{\widehat{\widehat{A}}_u}\, d\Gamma_u \,\Big|\, \mathcal{F}_t \right] \;=\; Q_1 + Q_2, \tag{26}$$

where (using $D_{uT_p}/\widehat{\widehat{D}}_{uT_p} = e^{\Gamma_{T_p} - \Gamma_u}$ under DI):

$$Q_1 \;=\; \mathbf{1}_{\tau>t}\, \alpha\, \widehat{\widehat{A}}_t\, R_t^{\mathrm{cmt}}\, e^{\Gamma_{T_p} - \Gamma_{T_s}} \big(1 - e^{-(\Gamma_{T_s} - \Gamma_t)}\big), \tag{27}$$

$$Q_2 \;=\; \mathbf{1}_{\tau>t}\, \beta_{T_p}\, e^{\Gamma_{T_p} - \Gamma_{T_s} + \Gamma_t}\, \widehat{\widehat{A}}_t\, \mathbb{E}^{\widehat{\widehat{A}}}\left[\int_t^{T_s} (R_u^{\mathrm{cmt}})^2\, e^{-\Gamma_u}\, d\Gamma_u \,\Big|\, \mathcal{F}_t \right]. \tag{28}$$

7 Lognormal CMT Closed-Form

7.1 Dynamics

Assumption 4 (Lognormal CMT)**.** *The CMT follows a driftless geometric Brownian motion under $\mathbb{Q}^{\widehat{\widehat{A}}}$:*

$$\frac{dR_t^{\mathrm{cmt}}}{R_t^{\mathrm{cmt}}} \;=\; \sigma\, dW_t^{\widehat{\widehat{A}}}, \tag{29}$$

with $\mathcal{F}$ generated by $W^{\widehat{\widehat{A}}}$, which remains a Brownian motion under the enlarged filtration $\mathcal{G}$.

7.2 Second moment and Q_2

Under (29), $\mathbb{E}^{\widehat{\widehat{A}}}[(R_u^{\mathrm{cmt}})^2 \mid \mathcal{F}_t] = (R_t^{\mathrm{cmt}})^2 e^{\sigma^2(u-t)}$. Hence:

$$\begin{aligned}\mathbb{E}^{\widehat{\widehat{A}}}\big[R_{T_s}^{\mathrm{cmt}}(\alpha + \beta_{T_p} R_{T_s}^{\mathrm{cmt}}) \,\big|\, \mathcal{F}_0\big] &= \alpha\, R_0^{\mathrm{cmt}} + \beta_{T_p}\,(R_0^{\mathrm{cmt}})^2\, e^{\sigma^2 T_s} \\ &= \frac{\chi_0\, R_0^{\mathrm{cmt}}\, \widehat{\widehat{D}}_{0T_p}}{\widehat{\widehat{A}}_0}\left[1 + \left(e^{\sigma^2 T_s} - 1\right)\left(1 - \frac{\alpha\, \widehat{\widehat{A}}_0}{\chi_0\, \widehat{\widehat{D}}_{0T_p}}\right)\right]. \end{aligned} \tag{30}$$

Constant intensity case

Under DI+AC with constant intensity γ (so $\Gamma_t = \gamma t$), (28) simplifies via Fubini and the lognormal second moment:

$$Q_2\big|_{t=0} = \chi_0\, e^{\Gamma_{T_p} - \Gamma_{T_s}}\, \widehat{\widehat{D}}_{0T_p}\left(1 - \frac{\alpha\, \widehat{\widehat{A}}_0}{\chi_0\, \widehat{\widehat{D}}_{0T_p}}\right) R_0^{\mathrm{cmt}}\, \frac{e^{(\sigma^2-\gamma)T_s} - 1}{\sigma^2 - \gamma}\, \gamma. \tag{31}$$

Stepwise intensity / volatility

For piecewise-constant (γ_i, σ_i) on a grid $u_0 = t, u_1, \ldots, u_n = T_s$,

$$Q_2 = \mathbf{1}_{\tau > t}\, \beta_{T_p}\, e^{\Gamma_{T_p} - \Gamma_{T_s}}\, (R_t^{\mathrm{cmt}})^2\, \widehat{\widehat{A}}_t\, I_t, \tag{32}$$

where

$$I_t = \sum_{i=1}^{n} \frac{e^{(\sigma_i^2-\gamma_i)u_i} - e^{(\sigma_i^2-\gamma_i)u_{i-1}}}{\sigma_i^2 - \gamma_i}\, \gamma_i\, e^{\Gamma_t - \Gamma_{u_{i-1}} - \sigma_i^2 t + \gamma_i u_{i-1}}. \tag{33}$$

7.3 Convexity correction formulas

Proposition 5 (CMT Convexity Corrections, lognormal case)**.** *Under DI and Assumption 4,*

$$\boxed{\mathrm{CC}_0^{(A)} = \mathrm{CC}_0^{(B)} = \left(e^{\sigma^2 T_s} - 1\right)\left(1 - \frac{\alpha\, \widehat{\widehat{A}}_0}{\chi_0\, \widehat{\widehat{D}}_{0T_p}}\right).} \tag{34}$$

Adding AC with constant intensity γ,

$$\boxed{\begin{aligned}\mathrm{CC}_0^{(C)} = \left(1 + \mathrm{CC}_0^{(B)}\right) \frac{\widehat{\widehat{D}}_{0T_s}}{D_{0T_s}} - 1 \\ + \frac{\alpha\, \widehat{\widehat{A}}_0}{\chi_0\, \widehat{\widehat{D}}_{0T_p}}\left(1 - e^{-\gamma T_s}\right) + \left(1 - \frac{\alpha\, \widehat{\widehat{A}}_0}{\chi_0\, \widehat{\widehat{D}}_{0T_p}}\right) \frac{e^{(\sigma^2-\gamma)T_s} - 1}{\sigma^2 - \gamma}\, \gamma.\end{aligned}} \tag{35}$$

Proof. For (A) and (B): the natural-discount-factor convention is $D \cdot F = \chi_0\, R_0^{\mathrm{cmt}}\, \widehat{\widehat{D}}_{0T_p} / \widehat{\widehat{A}}_0 \cdot \widehat{\widehat{A}}_0 = R_0^{\mathrm{cmt}}\, \chi_0\, \widehat{\widehat{D}}_{0T_p}$. Substituting (30) into Proposition 4 (24) gives (34) after using $\mathrm{NPV}^{(B)} = e^{\Gamma_{T_p} - \Gamma_{T_s}} \mathrm{NPV}^{(A)}$ and the matching denominator. For (C): split via Lemma 3, evaluate Q_1 and Q_2 by (27)–(31), normalise by $D_{0T_p} \cdot R_0^{\mathrm{cmt}}$ and rearrange. □

8 Limits and Sanity Checks

8.1 Vol-zero limit

As $\sigma \to 0$, $e^{\sigma^2 T_s} \to 1$, so $\mathrm{CC}^{(A)} = \mathrm{CC}^{(B)} = 0$. For (C), use $\widehat{\widehat{D}}_{0T_s}/D_{0T_s} = e^{-\gamma T_s}$ under DI+AC:

$$\mathrm{CC}_0^{(C)}\big|_{\sigma=0} \;=\; e^{-\gamma T_s} - 1 + \frac{\alpha \widehat{\widehat{A}}_0}{\chi_0 \widehat{\widehat{D}}_{0T_p}}(1 - e^{-\gamma T_s}) + \left(1 - \frac{\alpha \widehat{\widehat{A}}_0}{\chi_0 \widehat{\widehat{D}}_{0T_p}}\right)(1 - e^{-\gamma T_s}) \;=\; 0.$$

The cancellation works because the prefactor sums to 1 multiplied by $(1-e^{-\gamma T_s})$, exactly offsetting the leading $e^{-\gamma T_s} - 1$.

8.2 $\gamma \to \sigma^2$ limit

In (35), $(e^{(\sigma^2-\gamma)T_s} - 1)/(\sigma^2 - \gamma) \to T_s$ as $\gamma \to \sigma^2$, giving

$$\mathrm{CC}_0^{(C)}\big|_{\gamma=\sigma^2} \;\sim\; \left(1 + \mathrm{CC}_0^{(B)}\right) \frac{\widehat{\widehat{D}}_{0T_s}}{D_{0T_s}} - 1 \;+\; T_s\,\sigma^2. \tag{36}$$

Numerically the formula is well-behaved at the apparent singularity – use the Taylor expansion below in implementation.

8.3 No-default limit (Pelsser/Hagan)

In the limit $\gamma = 0$, $\widehat{\widehat{D}} = D$, $\widehat{\widehat{A}} = A_0$, $\chi_0 = 1$, and all three CCs collapse to

$$\boxed{\mathrm{CC} \;=\; \left(e^{\sigma^2 T_s} - 1\right)\left(1 - \alpha\,\frac{A_0}{D_{0T_p}}\right),} \tag{37}$$

which is exactly the Pelsser [7] CMS convexity formula under the same linear-swap-rate assumption. *This is the cleanest regression test against any classical CMS pricer.*

8.4 Vega sign

The vega $\partial \mathrm{CC}^{(A)}/\partial\sigma = 2\sigma T_s e^{\sigma^2 T_s}(1 - \alpha\widehat{\widehat{A}}_0/(\chi_0 \widehat{\widehat{D}}_{0T_p}))$. For $T_p \to T_s$, $\chi_0 \widehat{\widehat{D}}_{0T_p} > \alpha \widehat{\widehat{A}}_0$ (since $\alpha = 1/\sum y_i$ and the discounted annuity dominates a single discounted face value), so the vega is positive in the most common payment-date conventions.

9 Smile Extension via Static Replication

The lognormal closed forms can be extended to non-flat smile by replacing $\mathbb{E}^{\widehat{\widehat{A}}}[(R_{T_s}^{\mathrm{cmt}})^2]$ with its risky-smile-implied counterpart via Hagan's static replication [6]:

$$\mathbb{E}_0^{\widehat{\widehat{A}}}\left[(R_{T_s}^{\mathrm{cmt}})^2\right] \;=\; (R_0^{\mathrm{cmt}})^2 \;+\; 2\int_0^\infty C(K, T_s)\,dK, \tag{38}$$

where $C(K, T_s)$ is the price (per unit of $\widehat{\widehat{A}}_0$) of a risky CMS caplet at strike K and expiry T_s. This requires a risky CMS caplet smile – not directly market-quoted in most jurisdictions, so the practical route involves SABR interpolation of the swaption smile plus a credit-spread overlay.

10 Validation Notes for the Pricing Library

10.1 Risky-discount machinery

1. **Build** $D_{tT}, \widehat{D}_{tT}, \widehat{\widehat{D}}_{tT}$ from a risk-free curve plus a hazard curve. Under DI verify $\widehat{\widehat{D}}_{tT} = D_{tT}\, e^{\Gamma_t - \Gamma_T}$. For a flat hazard γ, $\widehat{D}_{0T} = e^{-\gamma T}\, D_{0T}$.
2. **Risky annuity.** Implement (10); smoke-test $\widehat{\widehat{A}}_0\, \widehat{\widehat{R}}_0^{\text{swp}} = \widehat{\widehat{D}}_{0T_s} - \widehat{\widehat{D}}_{0T_n}$ (recoveryless par-swap identity).
3. **CMT identification.** Use Proposition 2: implement R_0^{cmt} via (12) and verify it equals the level computed from the par-bond definition (3).

10.2 Linear model calibration

4. Verify $\alpha = 1/\sum y_i$ from Proposition 3. Compute each β_U via (21).
5. **Consistency identity.** Verify $\sum_i y_i\,(\alpha + \beta_{T_i} R_0^{\text{cmt}}) = \chi_0^{-1}$ (equivalent to $\sum_i y_i\, \widehat{\widehat{D}}_{0T_i}/\widehat{\widehat{A}}_0 = 1$).
6. **Lemma 4.** Implement $\mathbb{E}^{\widehat{\widehat{A}}}[\widehat{\widehat{D}}_{T_sU}/\widehat{\widehat{A}}_{T_s} \mid \mathcal{F}_0]$ via Monte Carlo and verify it equals $\chi_0\, \widehat{\widehat{D}}_{0U}/\widehat{\widehat{A}}_0$.

10.3 NPV identities

7. **(A)/(B) ratio.** Verify $\text{NPV}_0^{(B)}/\text{NPV}_0^{(A)} = e^{\Gamma_{T_p} - \Gamma_{T_s}}$. Under DI+AC with constant γ this is $e^{-\gamma(T_p - T_s)}$.
8. **Ordering.** Smoke test $\text{NPV}^{(C)} \geq \text{NPV}^{(B)} \geq \text{NPV}^{(A)}$.

10.4 Closed-form vs Monte Carlo

9. **Monte Carlo**: simulate R^{cmt} lognormally under $\mathbb{Q}^{\widehat{\widehat{A}}}$ on $[0, T_s]$; draw an independent default with intensity γ. Price (A), (B), (C) via the payoff definitions and compare against (34)–(35).
10. **No-default limit.** Set $\gamma = 0$ and verify all three CCs collapse to (37), the Pelsser CMS formula. This should match any classical CMS pricer to numerical precision.
11. **Vol-zero limit.** Set $\sigma = 0$; all three CCs must vanish (verify the algebraic cancellation in (35) using $\widehat{\widehat{D}}_{0T_s}/D_{0T_s} = e^{-\gamma T_s}$ under DI+AC).

10.5 Stepwise dynamics

12. Implement (32)–(33) for piecewise-constant (γ_i, σ_i). Validate against an MC with the corresponding piecewise dynamics.
13. **Numerical-stability tip.** For $|\sigma_i^2 - \gamma_i|$ near zero, replace $(e^{(\sigma_i^2 - \gamma_i)\,\Delta u} - 1)/(\sigma_i^2 - \gamma_i)$ with the Taylor expansion $\Delta u + \frac{1}{2}(\sigma_i^2 - \gamma_i)(\Delta u)^2 + \cdots$ to avoid catastrophic cancellation.

10.6 YTM–CMT bridge

14. For coupon $K \neq R_0^{\text{cmt}}$, use the Taylor link (4) to map a YTM-let into a CMT-let plus correction. Verify the linearisation error is $O((B-1)^2)$ for B near 1.
15. For an annual 30/360 bond, verify the closed form $R^{\text{ytm}}(B, K) \cong R^{\text{cmt}} + (K - R^{\text{cmt}}) - (B-1)/n$ holds at the level predicted by (4).

10.7 Smile extension (advanced)

16. **Static replication consistency.** Implement (38) and check it reproduces $(R_0^{\mathrm{cmt}})^2\, e^{\sigma^2 T_s}$ in the no-smile (Black) limit, then extends naturally to risky-smile inputs.

11 Connection to Other Pucci Papers

This paper is the credit-aware extension of the methodology used in the companion CMASW paper [8], sharing the same:

- *linear swap-rate assumption* (Hagan-style),
- *martingale exploitation* via choice of natural numeraire,
- *closed-form lognormal benchmark* that reduces to the classical Pelsser/Hagan formula in a deterministic limit.

The Index-Linked Hybrid paper [9] replaces the linear assumption with a full local-volatility specification, while the T-Lock paper [10] specialises to a deterministic-yield forward proxy.

Reference. Pucci, M. (2014). *Constant Maturity Treasury Convexity Correction.* International Journal of Theoretical and Applied Finance, Vol. 17, No. 8. SSRN no. 3387961.

Companion papers. M. Pucci (2012a), *Constant Maturity Asset Swap Convexity Correction*, SSRN 1961545 [8]; M. Pucci (2012b), *Pricing Index-Linked Hybrids under a Local Volatility Market Model*, SSRN 2056277 [9]; M. Pucci (2019), *Hedging the Treasury Lock*, SSRN 3386521 [10].

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